



MMA093 Rigid Body Dynamics

Assignment 5

Supplementary Part

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A. Time-and frequency-domain analysis

Problem 1: Vertical acceleration at the driver's seat

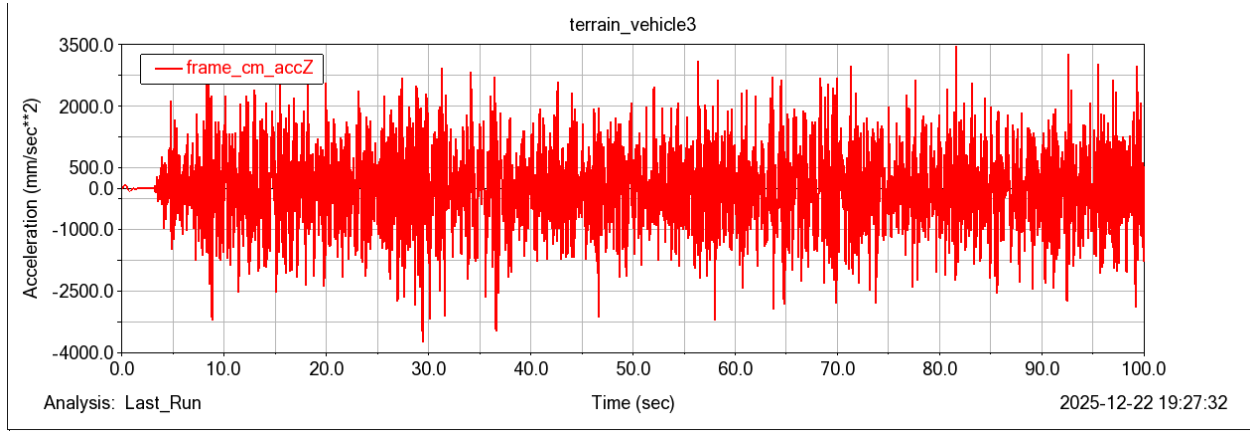


Figure 1: Vertical acceleration at the driver's seat

The time simulation is performed on the given road surface by running a dynamic simulation for 100 sec shown above fig 1.

Problem 2: Fast Fourier Transform

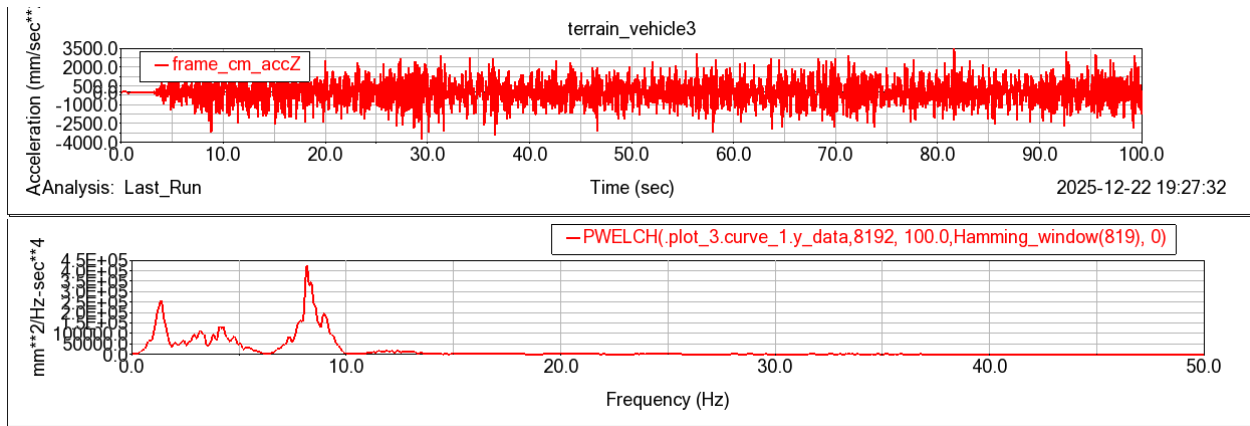


Figure 2: Frequency of the vertical acceleration

The frequency of the vertical acceleration is determined and two resonance is been located 1st peak is 1.416 Hz and 2nd peak is at 8.1421 Hz as shown above fig 2.

Problem 3: Eigen modes

The 62 and 65 are the two eigen modes that corresponds to the FFT frequency peaks. At eigen mode 62 we get 1.467 Hz and at eigen mode 65 we get 8.609 Hz which are closer to the value obtained in FFT.

Problem 4: Forced Vibration Analysis

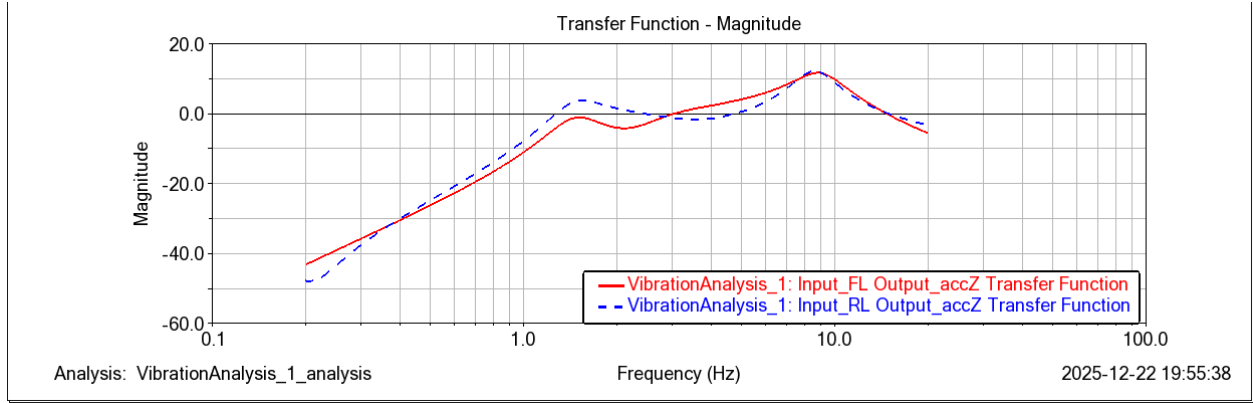


Figure 3: Front left and rear left input to the vertical acceleration

The simulation is done for 3 sec and two transfer function from the front left input and rear left input to the vertical acceleration output at the driver's seat is been plotted. Frequency resonance obtained are 1.4895 Hz and 8.571 Hz for the Front left and Rear left input as shown in above fig 3.

Problem 5: Matching resonance frequency to eigen mode

The resonance frequencies obtained in prob 4 are 1.4895 Hz and 8.571 Hz from the graph plotted and frequency obtained from non-linear model is 1.416 Hz and 8.609 Hz. The eigen mode which match to these frequencies are 62 and 65. At 62 eigen modes ATV vehicle is heave and pitch and at eigen mode 65 the front and rear wheel hop and very small amount of pitch due to rear wheel hopping at high resonance.

Problem 6: Decreased Damping coefficient

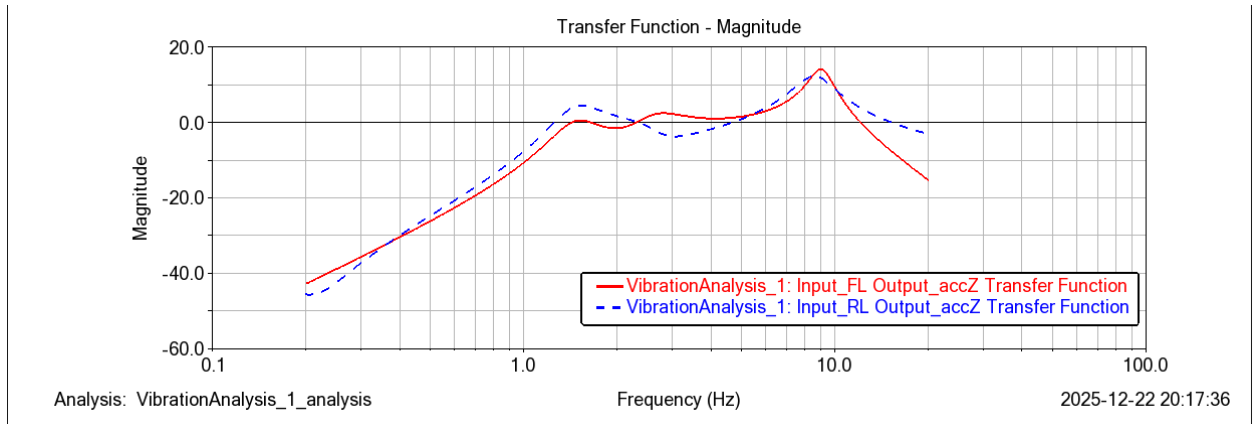


Figure 4: Transfer function for two damper values

The 1st small peak is occurred at 1.5172 Hz, 2nd peak is at 2.7608 Hz and 3rd peak is at 8.8926 Hz in the transfer function plot from front left and rear left input to vertical acceleration output as shown in above fig 4. The eigen modes which match to these frequencies are 64, 65 & 68 which has frequencies of 1.4473 Hz, 2.5602 Hz and 9.024 Hz and in eigen mode 64 ATV is heave condition and in 65 its pitching and in 68 the front wheel is hoping.

Problem 7: Increase Stiffness coefficient

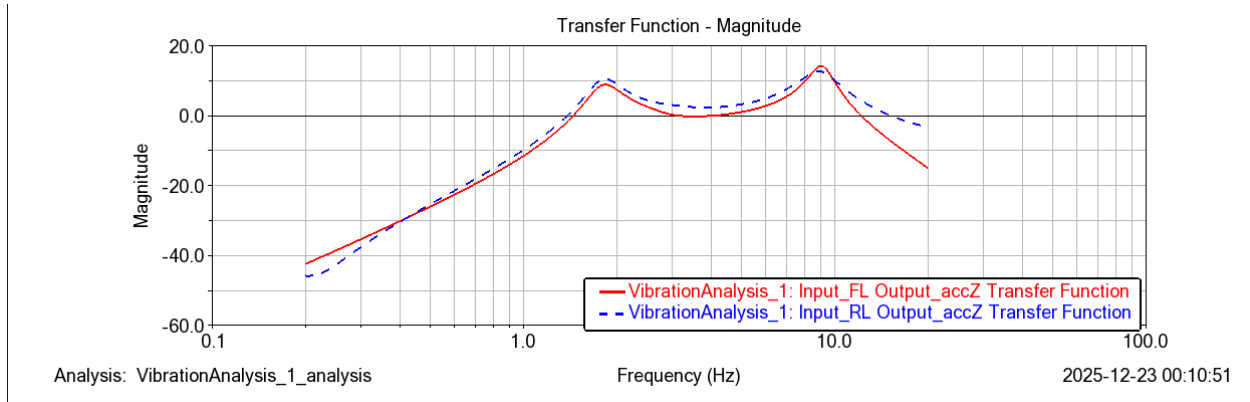


Figure 5: Transfer function from front and rear left input to vertical acceleration output

The 1st peak is occurred at 1.8073 Hz, 2nd peak is at 8.8926 Hz in the transfer function plot from front left and rear left input to vertical acceleration output as shown in above fig 5. The eigen modes which match to these frequencies are 63 & 66 which has frequencies of 1.8017 Hz and 8.8504 Hz and in eigen mode 63 ATV is heave condition and in eigen mode 66 the rear wheel is hoping and its slightly pitching.

Problem 8: Pros and cons

Time-Domain analysis

The Vehicle response is analysed as a function of time and Obtained directly from dynamic simulations.

Pros of Time domain

- Capture real vehicle behaviour which includes nonlinear suspension, damping and tire effects.
- Handles arbitrary inputs such as step inputs, bumps etc.
- Easy to see peaks, settling time, overshoot.
- Directly related to ride comfort and safety.
- Can be compared with experimental time signals.

Cons of Time domain

- Hard to identify resonance frequencies.

- Computationally expensive.
- A mode may not appear if not sufficiently excited.

Frequency-Domain Analysis

The vehicle response analyzed as a function of frequency using FFT, Transfer function, linearized eigen values.

Pros of Frequency Domain

- Natural frequencies and dominant modes are easy to locate.
- Efficient comparison can be made between Front vs rear input, different damper settings.
- Human sensitivity range (4–8 Hz). Standards often defined in frequency domain
- Each peak corresponds to a physical mode (heave, pitch, wheel hop).

Cons of Frequency domain

- Nonlinear effects are neglected.
- Does not show time history or transients.
- Not suitable for large disturbances such as severe off-road inputs.

B. Rear damper analysis

Problem 9: Linear Rear angle force and Swing-arm angle

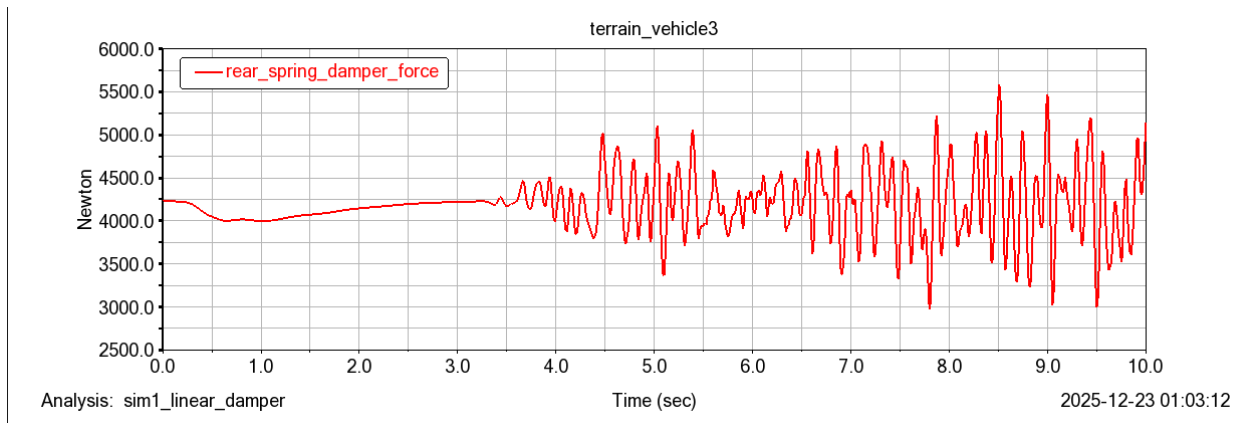


Figure 6: Rear spring damper force(N)

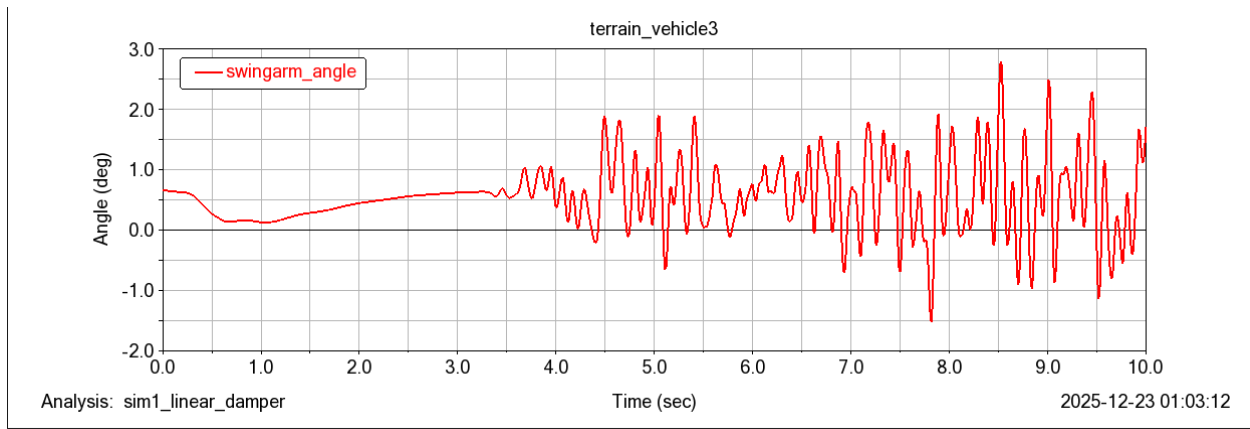


Figure 7: Swing-arm angle(deg)

The Rear damper force and swing-arm angle is been determined to the linear rear-damper.

Problem 10: Non-Linear Rear angle force and Swing-arm angle

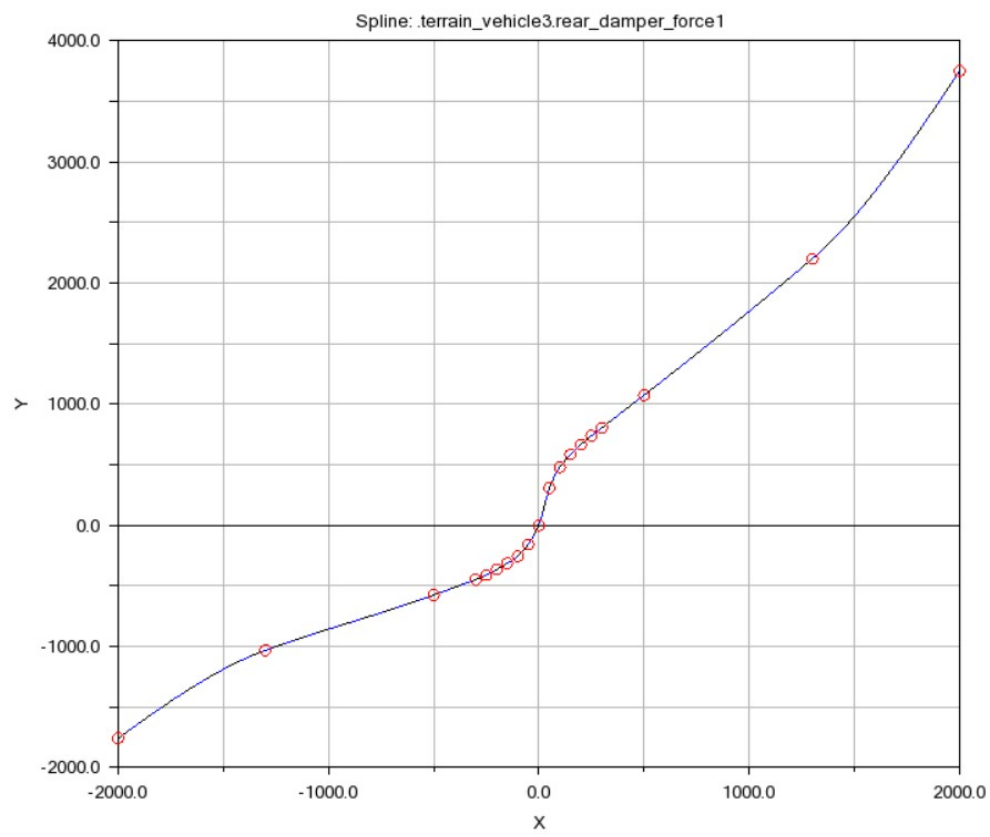


Figure 8: plotting data points and splines

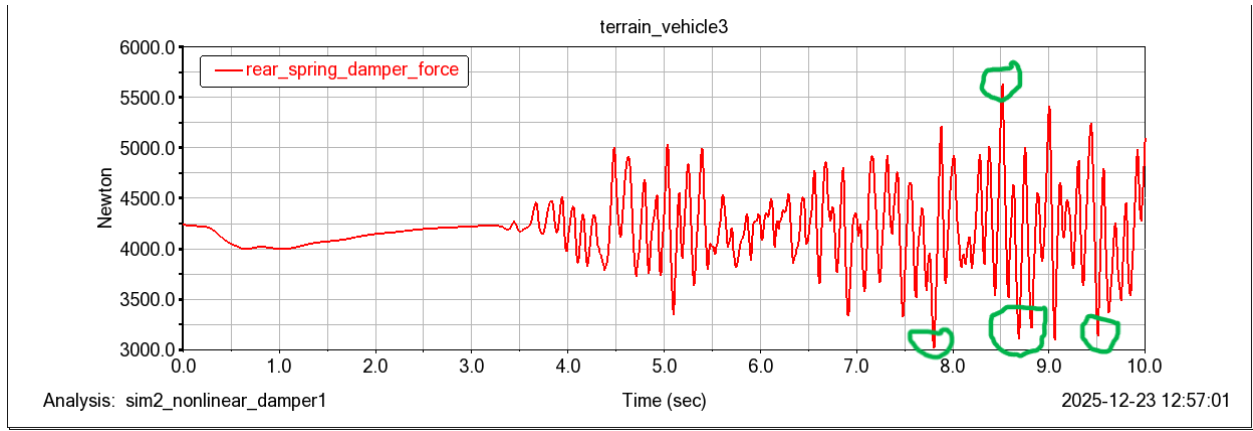


Figure 9: Rear spring damper force(N)

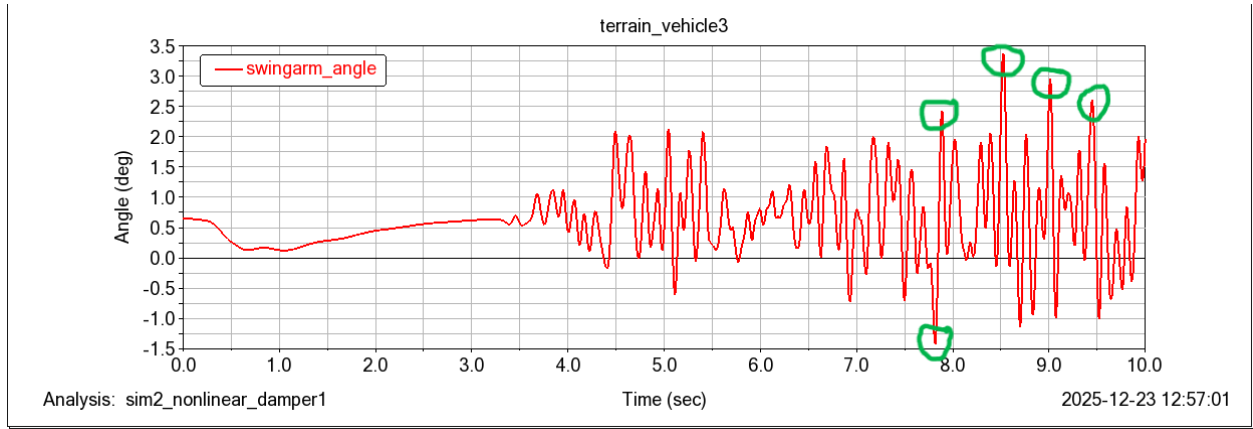


Figure 10: Swing-arm angle(deg)

The Rear damper force and swing-arm angle is been determined to nonlinear rear-damper data and it is been compared with the linear rear-damper obtained in problem 9.

The rear damper force upto 7.8 sec is similiar in both the cases but at beginning of 8 sec we can see higher peak forces in nonlinear case(marked in green) compared to the linear case. non linear damper data generated sharp and more irregular forces.

The swing-arm angle upto 7.8 sec is similar in both the cases and in the beginning of 8 sec we can see large angle amplitude oscillation in nonlinear damper data (marked in green) compared to linear case.

Problem 11: Road with a bump

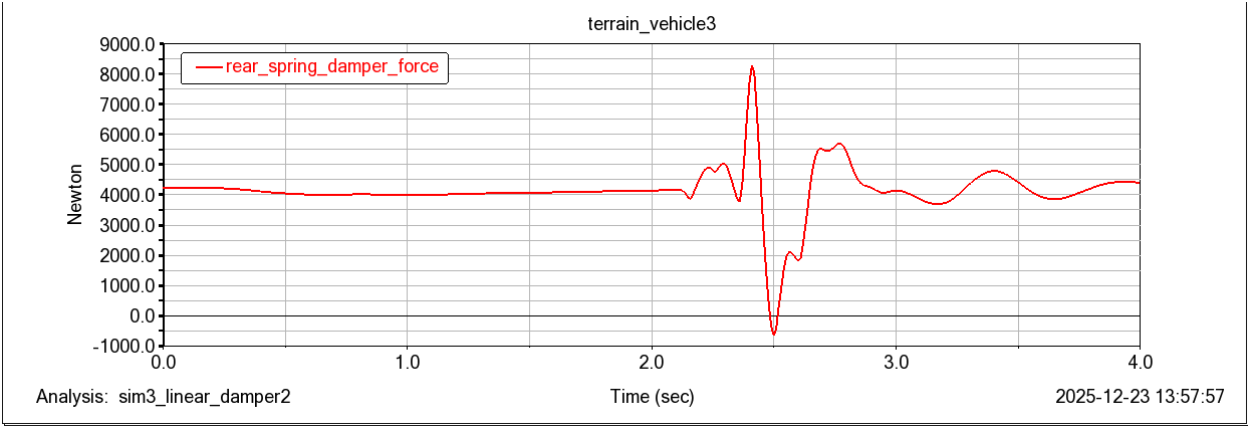


Figure 11: Linear Rear spring damper force(N)

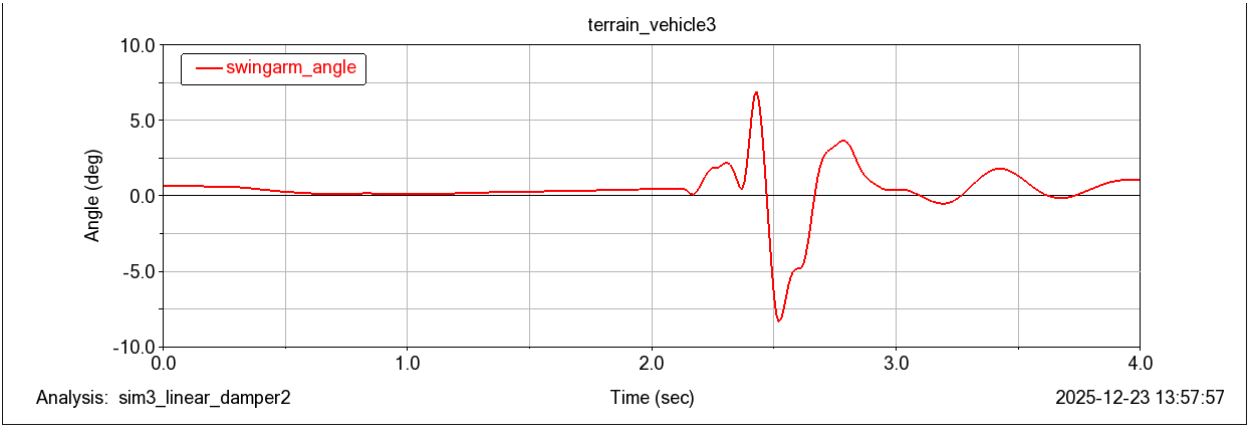


Figure 12: Linear swing-arm angle(deg)

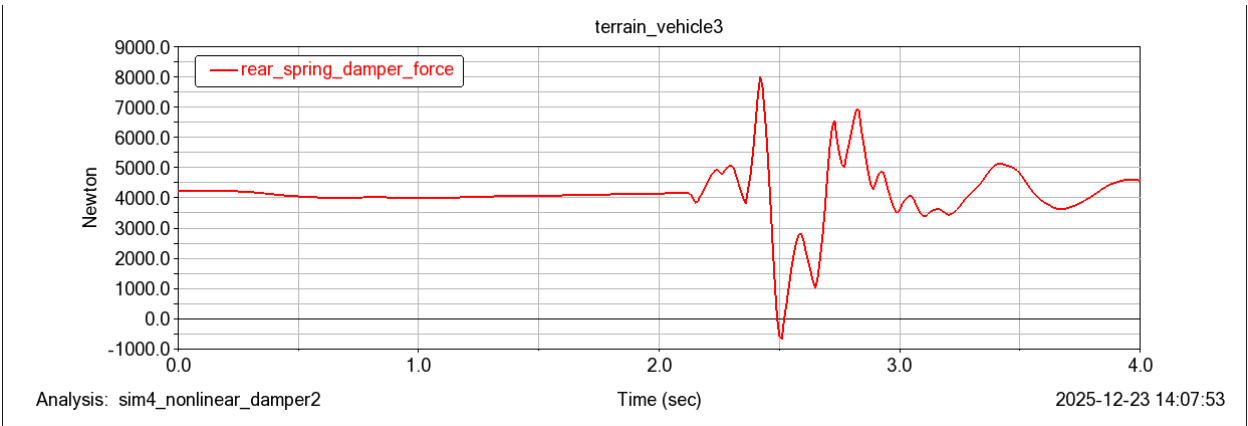


Figure 13: Nonlinear Rear spring damper force(N)

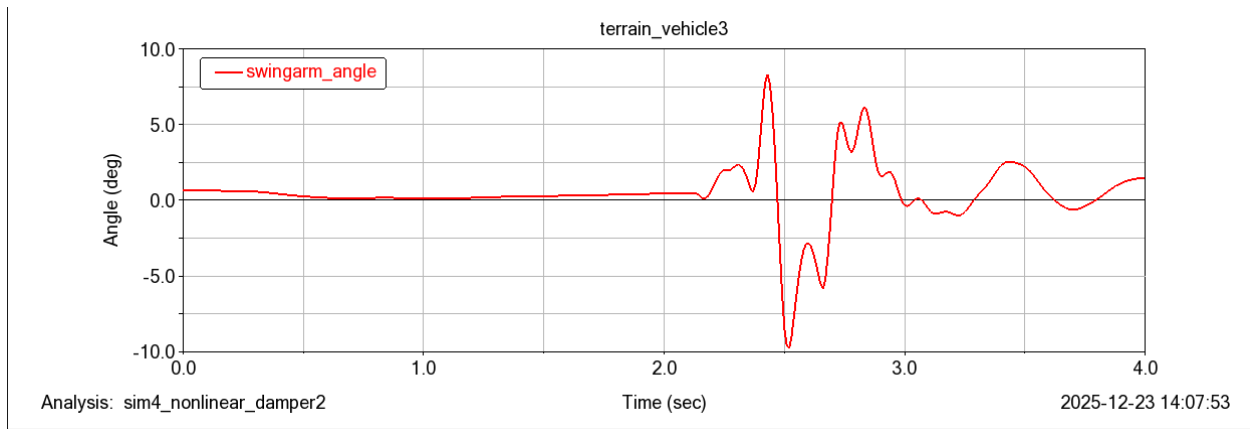


Figure 14: Nonlinear swing-arm angle(deg)

Problem 12: Road with a bump and nonlinear rear-damper data

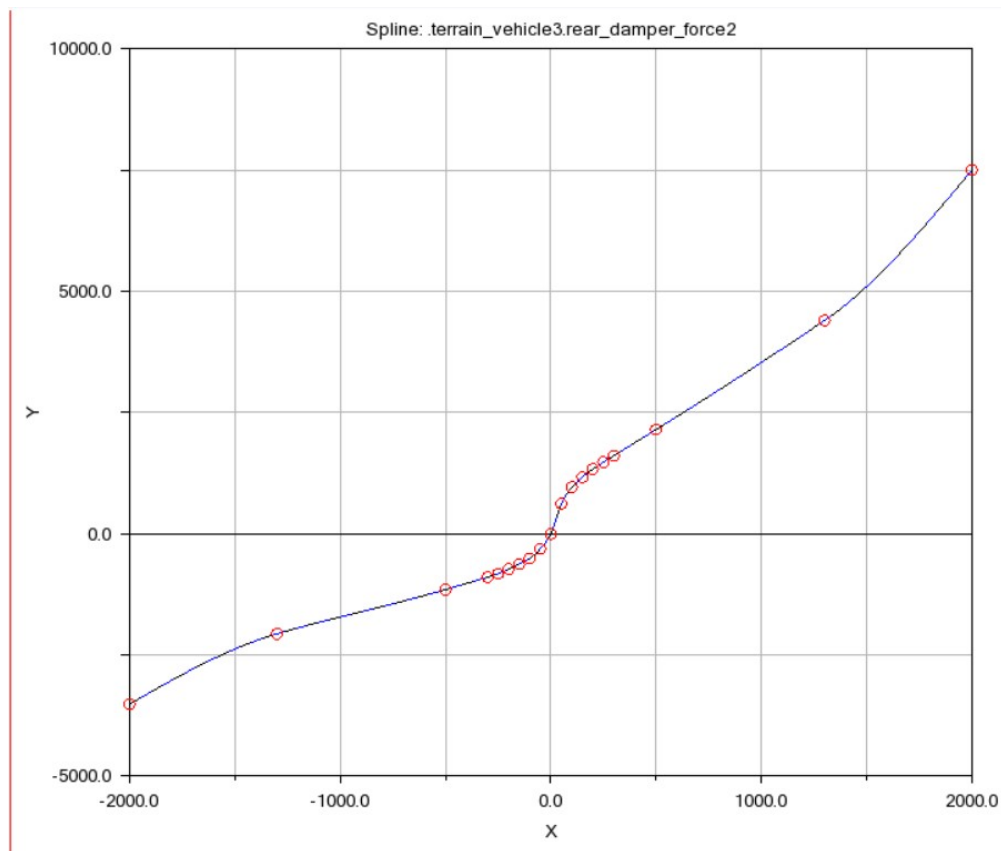


Figure 15: plotting data points and splines

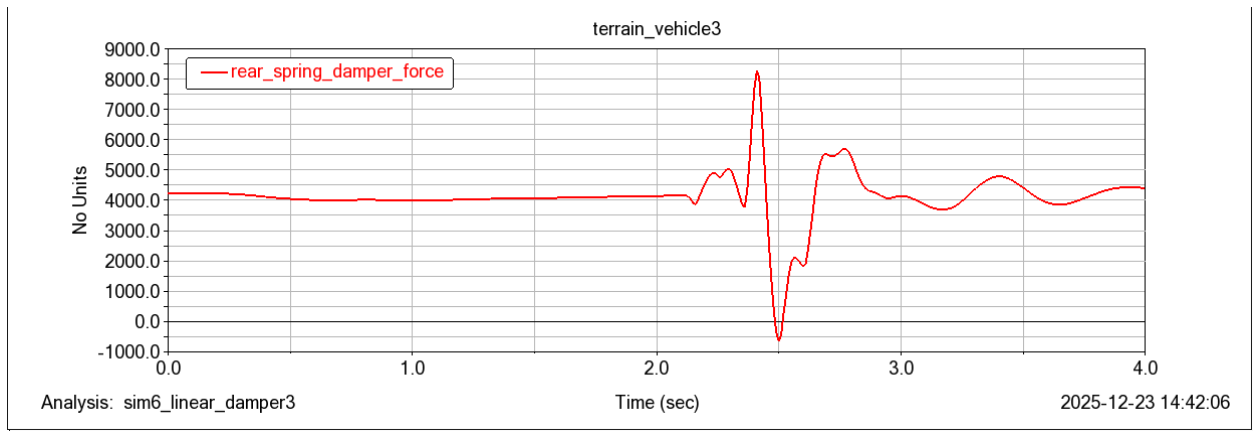


Figure 16: Linear Rear spring damper force(N)

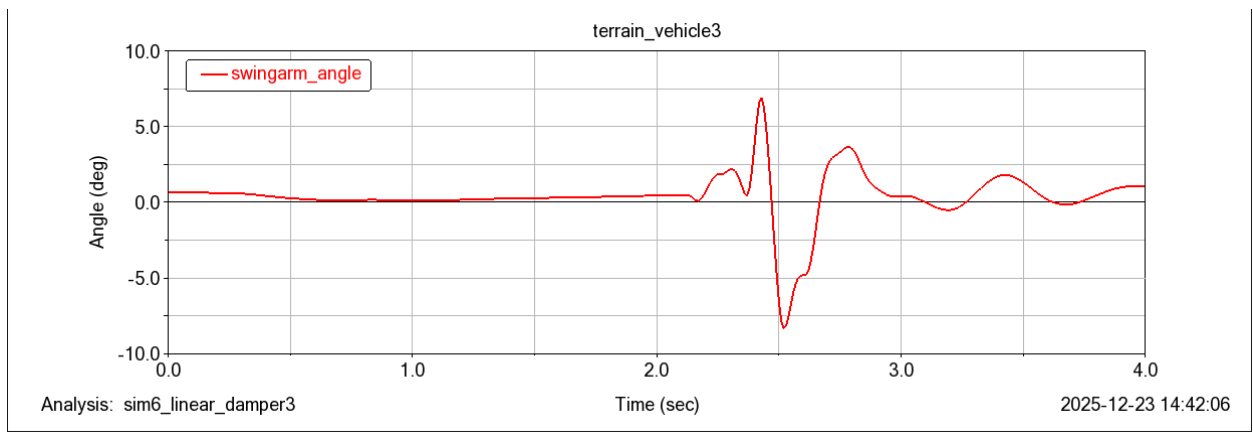


Figure 17: Linear swing-arm angle(deg)

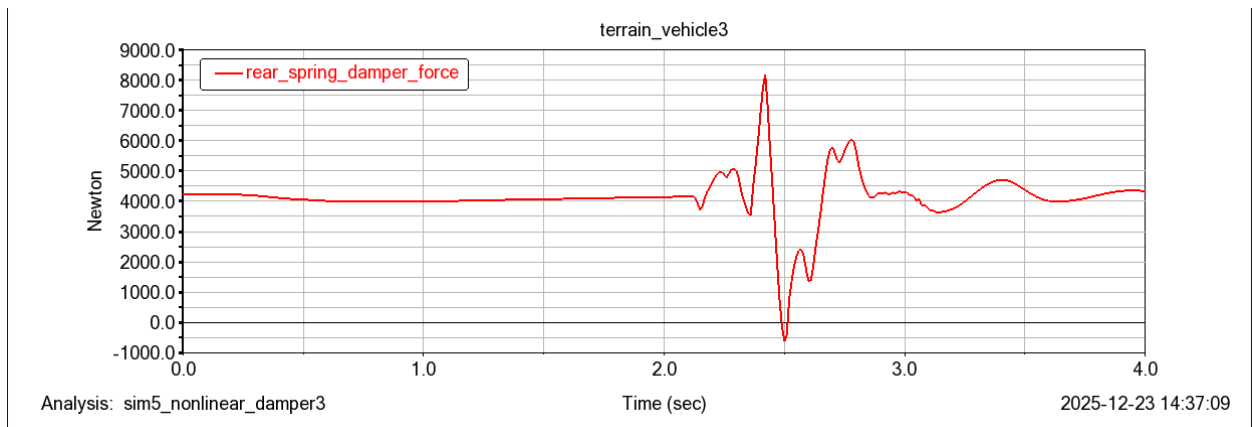


Figure 18: Nonlinear Rear spring damper force(N)

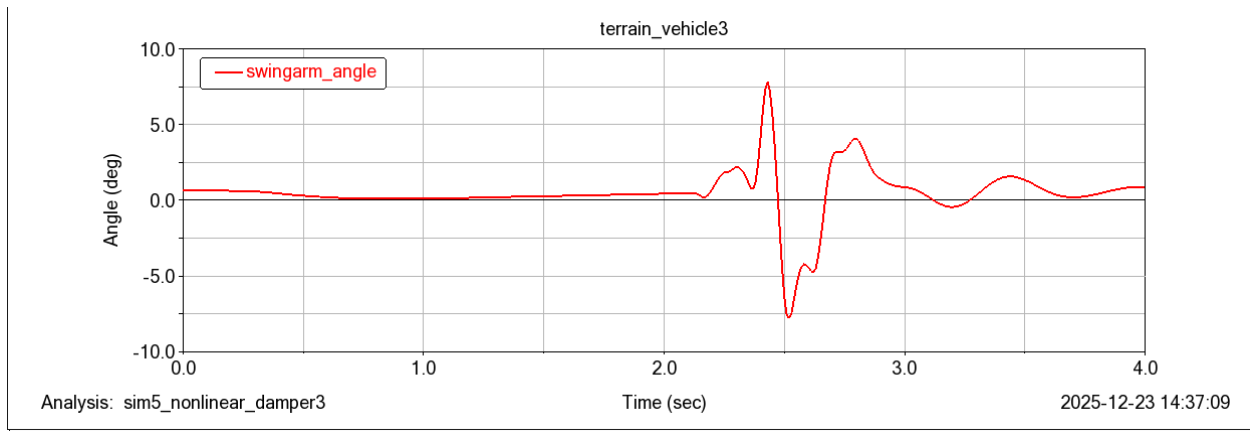


Figure 19: Nonlinear swing-arm angle(deg)

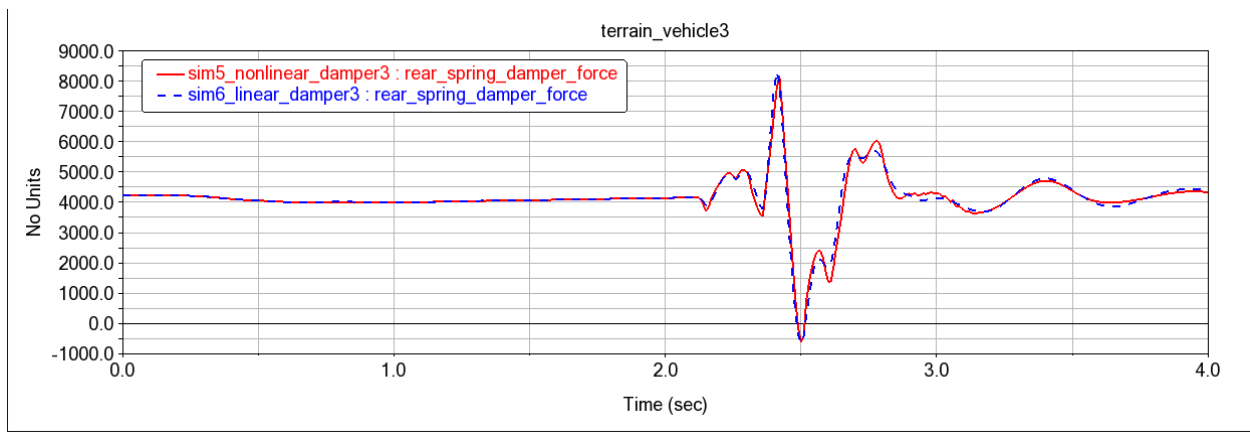


Figure 20: Copmarison of Rear spring damper force(N)

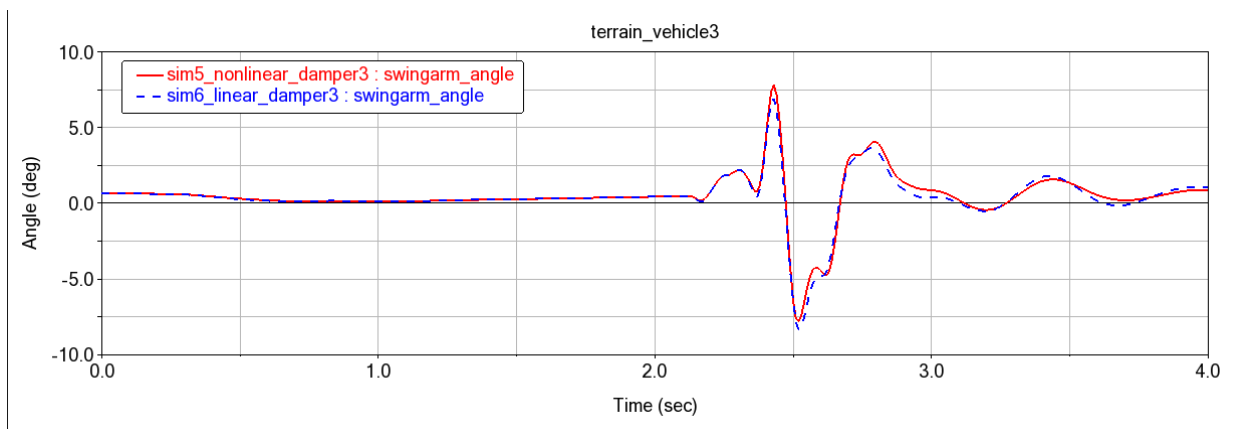
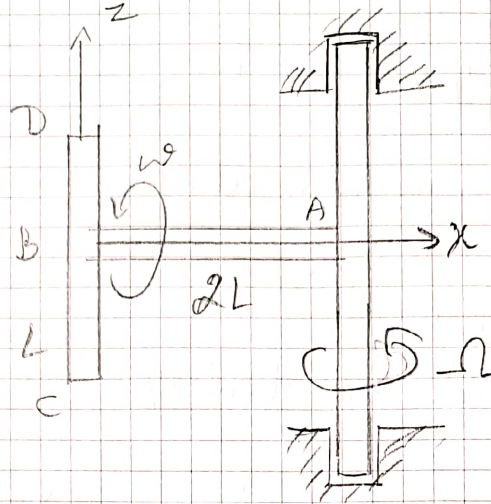


Figure 21: Comparison of swing-arm angle(deg)

For the given road with bump data and rear damper data2 is simulated and its results are compared with the linear and non linear rear spring damper force and swing-arm angle. For Rear spring damper force in fig 20 we can see that non linear has shaper peak forces and non uniform whereas linear has smooth and no sudden peak force. For Swing-arm angle in fig 21 we can see that the angle in the beginning is similar in both linear and non-linear and then non linear has slightly large angle amplitude then linear.

Problem 13: problem 2.6 in Bostrom.

Problem 13



Take origin at the vertical rotation axis through A.
 e_x along AB, e_z vertical & e_y perpendicular to.
 e_x .

The xyz system is only rotating with angular
velocity Ω .

$$\Omega = \Omega e_z$$

Rod AB lies along x -axis

Rod CD is vertical.

Rod length AB is $2L$
BC is L

Position vector

from B to C

$$\Rightarrow r_{C/B} = -Le_z.$$

From O to B

$$r_{B/O} = -L e_x$$

Rod CD has angular velocities.

① Rotation of frame: $\Omega = \Omega e_z$.

② Relative rotation CD about AB = $\omega = \omega e_x$

Total angular velocity $\omega_{CD} = \Omega e_z + \omega e_x$.

Velocity at Point C.

Vector differential formula.

$$V_C = V_B + \left(\frac{dr_{C/B}}{dt} \right)_{rot} + \Omega \times r_{C/B} \quad \text{--- (1)}$$

Velocity of point B

$$V_B = \Omega \times r_{B/O} = \Omega e_z \times (-L e_x) = -L \Omega e_y \quad \text{--- (2)}$$

Relative velocity of C wrt B

$$\left(\frac{dr_{C/B}}{dt} \right)_{rot} = \omega \times r_{C/B} = \omega e_x \times (-L e_z) = -L \omega e_y \quad \text{--- (3)}$$

Rotation frame due to transport term.

$$\Omega \times r_{C/B} = \Omega e_z \times (-L e_z) = 0 \quad \text{--- (4)}$$

Point C is not fixed to AB & AB itself rotates with Ω . Thus velocity of C has another Ω -contribution through B. This has another $-L\Omega e_y$. — (5)

Velocity of C. ~~do~~ substitute (2) (3) (4) & (5) in (1).

$$\boxed{V_C = -L(2\Omega + \omega) e_y}.$$

Acceleration of point C.

Rotating frame acceleration.

$$a_C = a_B + \left(\frac{d^2 r_{C/B}}{dt^2} \right)_{\text{rot}} + 2\Omega \times \left(\frac{dr_{C/B}}{dt} \right)_{\text{rot}} + \Omega \times (\Omega \times r_{C/B}) \quad \text{--- (6)}$$

Acceleration of point B

$$a_B = \Omega \times (\Omega \times r_{B/O}) = \Omega e_z \times (\Omega e_z \times (-L e_x))$$

$$a_B = 2L\Omega^2 e_x. \quad \text{--- (7)}$$

Relative acceleration due to ω

$$\alpha = \dot{\omega} e_x$$

$$\begin{aligned} \left(\frac{d^2 r_{C/B}}{dt^2} \right)_{\text{rot}} &= \alpha \times r_{C/B} + \omega \times (\omega \times r_{C/B}) \\ &= \dot{\omega} e_x \times (-L e_z) + \omega e_x \times (\omega e_x \times -L e_z) \\ &= -L\dot{\omega} e_y + L\omega^2 e_z. \quad \text{--- (8)} \end{aligned}$$

Acceleration

$$2\Omega \times \left(\frac{dr_{C/B}}{dt} \right)_{rot} = 2\Omega e_z \times (-L\omega e_y) = 2L\Omega\omega e_x \quad (9)$$

Frame Antijetw.

$$\Omega \times (\Omega \times r_{C/B}) = \Omega e_z \times (\Omega e_z \times -Le_z) = 0 \quad (10)$$

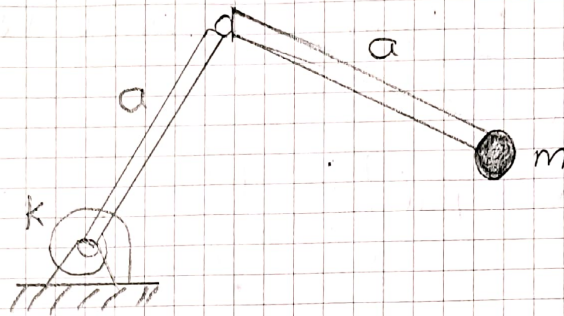
Acceleration at point C

Substitute (7) (8) (9) (10) in (6).

$$a_c = 2L\Omega(\Omega + \omega)e_x - L\omega e_y + L\omega^2 e_z$$

Problem 14: problem 6.5 in Bostrom.

Problem 14



Two massless rods, each of length a , connected by pin. Lower rod is pinned to ground & connected to a torsional spring of stiffness k .

Upper rod carries a point mass m at its free end.

Lower rod makes an angle $\vartheta = 30^\circ$ with horizontal

Generalized co-ordinates

q_1 = rotation of lower rod

q_2 = relative rotation of the upper rod

Linearised kinetic energy

Velocity of mass comes from the motion of both rods

$$T = \frac{1}{2} m a^2 (3 \dot{q}_1^2 + 2 \dot{q}_1 \dot{q}_2 + \dot{q}_2^2)$$

~~M~~ m

Position of mass m .

$$r_m = a \begin{bmatrix} \cos \theta_1 \\ \sin \theta_1 \end{bmatrix} + a \begin{bmatrix} \cos(\theta_1 + \theta_2) \\ \sin(\theta_1 + \theta_2) \end{bmatrix}$$

Mass Matrix

$$M = ma^2 \begin{bmatrix} 3 & 1 \\ 1 & 1 \end{bmatrix} \quad - (1)$$

Potential Energy

Torsional spring $V_k = \frac{1}{2} k q_1^2$

Gravity

$$\theta_a = 30^\circ$$

$$V_g = \frac{1}{2} mga (2q_1^2 + 2q_1 q_2 + q_2^2)$$

Total potential Energy

$$V = \frac{1}{2} [q_1 \ q_2] k \begin{bmatrix} q_1 \\ q_2 \end{bmatrix}$$

$$k = \begin{bmatrix} k + 2mga & mga \\ mga & mga \end{bmatrix} \quad - (2)$$

Linearised Equation of Motion.

$$\det(k - \omega^2 M) = 0. \quad - (3)$$

Substitute (1) & (2) in (3)

$$\begin{bmatrix} k + 2mga - 3ma^2\omega^2 & mga - ma^2\omega^2 \\ mga - ma^2\omega^2 & mga - ma^2\omega^2 \end{bmatrix} = 0$$

$$\lambda = \frac{a\omega^2}{g} \quad \alpha = \frac{2k}{mga} \quad \text{--- (4)}$$

Divide determinant by mga.

$$\lambda^2 - \frac{1}{3}(1+\alpha)\lambda - \frac{1}{3}(1-\alpha) = 0.$$

$$\lambda = \frac{1}{6} \left[(1+\alpha) \pm \sqrt{(1+\alpha)^2 + 6(1-\alpha)} \right] \quad \text{--- (5)}$$

Substitute (4) in (5)

$$\omega^2 = \frac{g}{6a} \left[(1+\alpha) \pm \sqrt{(1+\alpha)^2 + 6(1-\alpha)} \right]$$

Divide Multiply Numerator & denominator by 2

$$\omega = \sqrt{\frac{g}{3a} \left[1 + \frac{2k}{mga} \pm \sqrt{\left(1 + \frac{2k}{mga}\right)^2 + 6\left(1 - \frac{2k}{mga}\right)} \right]}$$

$$\omega = \left(\frac{g}{3a} \left[1 + \frac{2k}{mga} \pm \sqrt{\left(1 + \frac{2k}{mga}\right)^2 + 6\left(1 - \frac{2k}{mga}\right)} \right] \right)^{1/2}$$